Experiment 9.2

Simulation of Space Wave Propagation Characteristics using MATLAB

Aim

To simulate space wave propagation and to calculate free space path loss, line-of-sight (LOS) distance, received power and link budget.

Software Required

MATLAB (MATLAB Online)

Procedure / Algorithm

- Take f = 900 MHz, transmit power Pt = 30 dBm and distance d = 1 to 100 km.
- Calculate FSPL = 32.44 + 20log$_{10}$(f) + 20log$_{10}$(d) and received power Pr = Pt – FSPL.
- Take ht = 50 m, hr = 25 m and calculate LOS distance = 4.12(√ht + √hr).
- Do the link budget with Gt = 10 dB and Gr = 8 dB, and plot the results.

Output

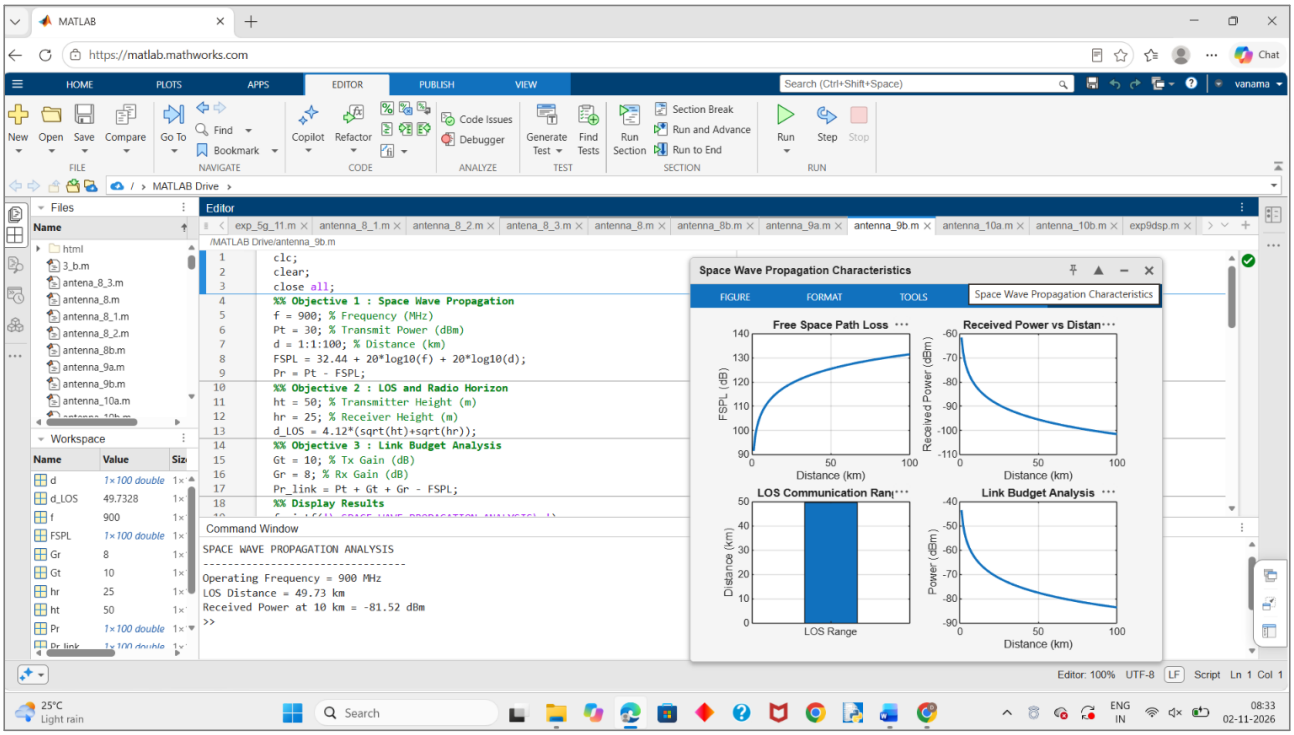


Fig. 9.2 Space wave propagation characteristics

Result

- LOS distance = 49.73 km at 900 MHz.
- Received power at 10 km = –81.52 dBm.
- Path loss increases and received power decreases with distance, as expected.